

WildGuard India

Wildlife Conservation Analytics & Risk Dashboard

Project Report

Data Source: Living Planet Index 2024 Public Data

India Subset Analysis

1. Abstract / Executive Summary

WildGuard India is a data analytics project that analyses wildlife population trends in India using the Living Planet Index (LPI) 2024 public dataset. The project cleans and processes 280 population time-series covering 127 species across 5 taxonomic classes, computes scientifically meaningful trend metrics using log-linear regression, and presents findings through an interactive Streamlit dashboard.

Key findings include: 25.0% of monitored population series show declining trends, with a median annual change of -3.73%. The dashboard identifies high-risk populations, reveals patterns across taxonomic groups and ecosystems, and provides evidence-based conservation recommendations.

2. Problem Statement

India harbours exceptional biodiversity but faces mounting conservation challenges from habitat loss, pollution, climate change, and human-wildlife conflict. Effective resource allocation requires understanding which species and ecosystems are most at risk.

Central question: "What do India's wildlife population trends tell us about conservation risk, and where should conservation attention be prioritised?"

3. Objectives

- Clean and prepare the India subset of the LPI dataset for analysis.
- Compute population trend metrics using scientifically appropriate methods.
- Identify declining, increasing, stable, and uncertain population series.
- Quantify conservation risk through meaningful KPIs.
- Present findings in an accessible, interactive dashboard.
- Generate actionable conservation insights.

4. Dataset Description

Source: Living Planet Index 2024 Public Data
URL: https://www.livingplanetindex.org/data_portal
File: LPD_2024_public.csv

The full dataset contains 35,996 population time-series from around the world. The India subset (Country == 'India', Included in LPR2024 == 1) contains 280 series after quality filtering.

Key Characteristics:

Population series:	280
Unique species:	127
Taxonomic classes:	5 (Aves, Mammalia, Reptilia, Actinopteri, Elasmobranchii)
Ecosystems:	3 (Terrestrial, Freshwater, Marine)

Observation period: 1956 - 2018

5. Data Cleaning

The following cleaning steps were applied:

- Filtered to Country == 'India' (exact match; excludes 'British Indian Ocean Territory').
- Retained only series with Included in LPR2024 == 1 (non-replicate, quality-checked).
- Converted year columns to numeric; coerced non-numeric/NULL values to NaN.
- Removed series with fewer than 2 valid positive observations.
- Stripped whitespace from categorical fields.
- Dropped empty year columns and trailing unnamed columns.
- No random sampling was applied -- all qualifying India records are preserved.

Records after cleaning: 280

6. Exploratory Data Analysis

Taxonomic Distribution:

Aves (birds) is the most represented class, followed by Mammalia. Reptilia, Actinopteri (ray-finned fish), and Elasmobranchii (sharks/rays) have smaller representation.

Ecosystem Distribution:

Freshwater and Terrestrial ecosystems dominate the India LPI data. Marine species are relatively under-represented.

Trend Classification:

Declining:	70 (25.0%)
Increasing:	65 (23.2%)
Stable:	6 (2.1%)
Uncertain:	139 (49.6%)

7. KPI Selection

Six KPIs were selected based on the actual data:

1. Population Series Count:	280
2. Unique Species Count:	127
3. Declining Series Count:	70
4. Percentage Declining:	25.0%
5. Median Annual Change:	-3.73%
6. Taxonomic Classes Covered:	5

8. Methodology

Population Trend Calculation:

LPI population series use heterogeneous abundance measures (counts, density, indices, encounter rates) that cannot be meaningfully summed across species. Instead, trends are computed using log-linear regression: $\ln(\text{population}) \sim \text{year}$.

The slope gives the mean proportional rate of change per year. Annualised % change = $(\exp(\text{slope}) - 1) \times 100$. This approach handles different measurement units correctly since relative change is unit-free.

Trend Classification:

- Declining: slope < -0.01 and $p < 0.1$
- Increasing: slope $> +0.01$ and $p < 0.1$
- Stable: $|\text{slope}| \leq 0.01$ and $p < 0.1$
- Uncertain: $p \geq 0.1$ (insufficient statistical evidence)

Risk Identification:

High-risk populations are identified as those with Declining trend classification AND annual percentage change $< -3\%$. A total of 68 series meet this threshold.

9. Dashboard Description

Page 1 - Executive Overview:

- 6 KPI cards (series count, species, declining count, % declining, median change, classes)
- Aggregate wildlife population index (line chart)
- Trend classification breakdown (donut chart)
- 3 automatically derived insights

Page 2 - Wildlife Trends:

- Trend distribution by taxonomic class (grouped bar chart)
- Annual % change distribution (histogram) and by ecosystem (box plot)
- Top 15 declining and top 15 increasing population series
- Species coverage by class

Page 3 - Conservation Risk & Insights:

- High-risk population series table
- Risk indicators by class and ecosystem (bar charts)
- Geographic map of monitored populations
- 5 structured insights (FACT -> INSIGHT -> RISK/OPPORTUNITY -> ACTION)
- Recommended conservation actions and limitations

10. Key Findings

- 25.0% of India's monitored wildlife population series show declining trends.
- Aves is the most monitored class; this may reflect monitoring bias.
- Freshwater ecosystems account for a large share of monitoring effort.
- The median annual change is -3.73%, suggesting overall negative trajectory.
- 68 high-risk populations (annual decline > 3%) span multiple groups and ecosystems.
- Several populations show recovery, demonstrating conservation can work.
- Data sparsity is a major limitation -- many series have only 2-3 data points.

11. Conservation Risks

- Populations with sustained annual declines exceeding 3% face risk of local collapse.
- Under-monitored groups (fish, reptiles) may harbour undetected declines.
- Freshwater ecosystems face increasing threats from pollution, damming, and extraction.
- Temporal gaps mean current status is unknown for some historically monitored populations.

12. Opportunities

- Population recovery stories provide models for effective conservation.
- India's protected area network can be leveraged to expand monitoring.
- Citizen science data could help fill geographic and taxonomic gaps.
- Cross-border conservation partnerships could amplify impact.

13. Recommended Actions

- Strengthen monitoring: Expand long-term monitoring for under-surveyed groups.
- Protect critical habitats: Focus on regions with highest declining species concentration.
- Targeted species programmes: Develop recovery plans for steepest-declining populations.
- Replicate successes: Study and scale strategies associated with increasing trends.
- Update outdated surveys: Re-survey populations with data gaps exceeding 10 years.

14. Limitations

- All trends are observational and associational; causation cannot be inferred.
- Sampling is biased toward well-studied taxa (birds, large mammals).
- Different series use different abundance measures; trends use relative change.
- Many series have very few data points, limiting trend precision.
- Geographic coverage is uneven across India.
- No threat or driver information is included in the dataset.

- Some series have most recent observations before 2000.

15. Conclusion

WildGuard India demonstrates that meaningful conservation insights can be extracted from publicly available population monitoring data. The analysis reveals that a significant proportion of India's monitored wildlife populations show declining trends, with the steepest declines concentrated in specific taxonomic groups and ecosystems. At the same time, population recoveries in some species illustrate the potential for effective conservation action.

The dashboard provides a practical tool for prioritising conservation attention and tracking India's wildlife population health over time.

16. Dataset Source

Living Planet Index 2024 Public Data

https://www.livingplanetindex.org/data_portal

Citation: LPI 2024. Living Planet Index database. 2024. www.livingplanetindex.org